Veith Weilnhammer

Curriculum Vitae

Max Planck UCL Centre for Computational Psychiatry and Ageing Research, London, UK
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Personal Statement

I am a neuroscientist, psychiatrist, and psychotherapist. I work on digital technologies to better understand, predict, and improve mental health.

Education and Degrees

2023 **Venia legendi (Habilitation) in Experimental Psychiatry**, Charité, Berlin, Germany.
The construction of unambiguous conscious experiences from ambiguous sensory information.

2015 **Doctorate in Medicine (Dr. med.)**, Charité, Berlin, Germany.
Frontoparietal cortex mediates perceptual transitions in bistable perception (summa cum laude).

2014 **Medical degree and final medical examination**, Charité, Berlin, Germany.

2007–2014 **Medical studies**, Charité, Berlin, Germany, and Universidade Nova, Lisbon, Portugal.

2006–2007 **Studium Generale**, Leibniz Kolleg, Eberhard Karls Universität, Tübingen, Germany.

Licensure and Registration

2026 **Specialist Registration in Psychiatry**, General Medical Council, United Kingdom.

2023 **Specialist Qualification in Psychiatry & Psychotherapy**, Berlin Medical Association, Germany.

Professional Experience

2026–present **Clinician Scientist**, Copilot Health, Microsoft AI (contract), London.

2026–present **Honorary Consultant Psychiatrist**, Queen Square Neuropsychiatry, University College London Hospitals NHS Foundation Trust, London.

2025–present **Max Planck Fellow**, Max Planck UCL Centre for Computational Psychiatry and Ageing Research, London.

2023–2025 **Leopoldina Fellow**, Helen Wills Neuroscience Institute, University of California, Berkeley.

2015–2023 **Clinician Scientist**, Department of Psychiatry, Charité, Berlin.

2016–2023 **Psychotherapist in Training**, Centre for Psychotherapy, Humboldt-Universität, Berlin.

2015–2023 **Psychiatrist in Training**, Department of Psychiatry, Charité, Berlin.

2012–2013 **Minerva Fellow**, Weizmann Institute, Rehovot.

Fellowships and Awards

2027–2029 **NARSAD Young Investigator Award**, Brain & Behavior Research Foundation

2025–2026 **NARSAD Young Investigator Award**, Brain & Behavior Research Foundation

2023–2025 **Leopoldina Postdoctoral Fellowship**, German National Academy of Sciences

2023 **Walter Benjamin Fellowship**, DFG (awarded; declined in favor of Leopoldina)

2019–2022 **Clinician Scientist Fellowship**, Berlin Institute of Health

2017–2018 **Junior Clinician Scientist Fellowship**, Berlin Institute of Health

2013 **Minerva Short Term Fellowship**, Minerva Foundation, Max Planck Society

2007–2015 **University Scholarship**, German Academic Scholarship Foundation

Key Publications

2026 **Veith Weilnhammer**, Kevin YC Hou, Lennart Luettgau, Christopher Summerfield, Raymond Dolan, and Matthew M. Nour. A clinically validated framework for auditing AI chatbot behavior in mental health interactions. *Nature Medicine*, August 2026.

2025 **Veith Weilnhammer**, Marcus Rothkirch, Deniz Yilmaz, Merve Fritsch, Lena Esther Ptaszynski, Katrin Reichenbach, Lukas Rödiger, Philip Corlett, and Philipp Sterzer. N-methyl-d-aspartate receptor hypofunction causes recurrent and transient failures of perceptual inference. *Brain*, volume 148(5), pages 1531–1539, May 2025.

2025 **Veith Weilnhammer**, Jefferson Ortega, and David Whitney. Human-computer interactions predict mental health. *arXiv*, page 2511.20179 [q-bio], November 2025.

2024 **Veith Weilnhammer**, Yuki Murai, and David Whitney. Dynamic predictive templates in perception. *Current Biology*, volume 34(18), pages 4301–4306.e2, September 2024.

2021 **Veith Weilnhammer**, Merve Fritsch, Meera Chikermane, Anna-Lena Eckert, Katharina Kanthak, Heiner Stuke, Jakob Kaminski, and Philipp Sterzer. An active role of inferior frontal cortex in conscious experience. *Current Biology*, volume 31(13), pages 2868–2880.e8, July 2021.

Peer-Reviewed Publications

2026 **Veith Weilnhammer**, Kevin YC Hou, Lennart Luettgau, Christopher Summerfield, Raymond Dolan, and Matthew M. Nour. A clinically validated framework for auditing AI chatbot behavior in mental health interactions. *Nature Medicine*, August 2026.

2026 Isabella Goodwin, Kelly M. J. Diederen, Emily J. Hird, **Veith Weilnhammer**, Marta I. Garrido, and Franziska Knolle. Predictive processing accounts of psychosis: bottom-up or top-down disruptions. *Nature Mental Health*, volume 4(1), pages 60–84, January 2026.

2026 Merve Fritsch, Jochen Michely, Lucca Jaeckel, Ida Rangus, Christoph Riegler, Jan Friedrich Scheitz, Christian Nolte, Philipp Sterzer, and **Veith Weilnhammer**. Ischemic lesions to the inferior frontal cortex slow perceptual switching. *iScience*, volume 29(3), page 114943, March 2026.

2025 **Veith Weilnhammer**, Marcus Rothkirch, Deniz Yilmaz, Merve Fritsch, Lena Esther Ptaszynski, Katrin Reichenbach, Lukas Rödiger, Philip Corlett, and Philipp Sterzer. N-methyl-d-aspartate receptor hypofunction causes recurrent and transient failures of perceptual inference. *Brain*, volume 148(5), pages 1531–1539, May 2025.

2025 Anna-Chiara Schaub, Anna-Lena Eckert, Stijn A. Nuiten, **Veith Weilnhammer**, and Philipp Sterzer. Reduced weighting of short-term perceptual priors during auditory perceptual decision-making in psychosis-prone individuals. *BMC Biology*, volume 23(1), page 310, October 2025.

2025 Katrin Reichenbach, Marcus Rothkirch, Lucca Jaeckel, Philipp Sterzer, and **Veith Weilnhammer**. Anterior insular activity signals perceptual conflicts induced by temporal and spatial context. *Neuroscience of Consciousness*, volume 2025(1), page niaf030, 2025.

2024 **Veith Weilnhammer**, Yuki Murai, and David Whitney. Dynamic predictive templates in perception. *Current Biology*, volume 34(18), pages 4301–4306.e2, September 2024.

2024 **Veith Weilnhammer**. Where is the ghost in the shell? *Neuroscience of Consciousness*, volume 2024(1), page niae015, January 2024.

2023 **Veith Weilnhammer**, Heiner Stuke, Kai Standvoss, and Philipp Sterzer. Sensory processing in humans and mice fluctuates between external and internal modes. *PLOS Biology*, volume 21(12), page e3002410, December 2023.

2023 Heiner Stuke, Kathlen Priebe, **Veith Weilnhammer**, Hannes Stuke, and Nikola Schoofs. Sparse models for predicting psychosocial impairments in patients with PTSD: An empirical Bayes approach. *Psychological Trauma: Theory, Research, Practice, and Policy*, volume 15(1), pages 80–87, January 2023.

2023 Laurie-Anne Sapey-Triomphe, Lauren Pattyn, **Veith Weilnhammer**, Philipp Sterzer, and Johan Wagemans. Neural correlates of hierarchical predictive processes in autistic adults. *Nature Communications*, volume 14(1), page 3640, June 2023.

2023 Merve Fritsch, **Veith Weilnhammer**, Paul Thiele, Andreas Heinz, and Philipp Sterzer. Sensory and environmental uncertainty in perceptual decision-making. *iScience*, volume 26(4), page 106412, April 2023.

2022 Laurie-Anne Sapey-Triomphe, **Veith Weilnhammer**, and Johan Wagemans. Associative learning under uncertainty in adults with autism: Intact learning of the cue-outcome contingency, but slower updating of priors. *Autism*, volume 26(5), pages 1216–1228, July 2022.

2022 Stephan Köhler, **Veith Weilnhammer**, Henrik Walter, Susanne Erk, Philipp Sterzer, and Anne Guhn. Autobiographical script-driven imagery has no detectable effect on emotion regulation in healthy individuals. *Neuropsychobiology*, volume 81(2), pages 141–148, 2022.

2021 **Veith Weilnhammer**, Merve Fritsch, Meera Chikermane, Anna-Lena Eckert, Katharina Kanthak, Heiner Stuke, Jakob Kaminski, and Philipp Sterzer. An active role of inferior frontal cortex in conscious experience. *Current Biology*, volume 31(13), pages 2868–2880.e8, July 2021.

2021 **Veith Weilnhammer**, Meera Chikermane, and Philipp Sterzer. Bistable perception alternates between internal and external modes of sensory processing. *iScience*, volume 24(3), page 102234, March 2021.

2021 Heiner Stuke, Elisabeth Kress, **Veith Weilnhammer**, Philipp Sterzer, and Katharina Schmack. Overly strong priors for socially meaningful visual signals are linked to psychosis proneness in healthy individuals. *Frontiers in Psychology*, volume 12, page 583637, May 2021.

2020 **Veith Weilnhammer**, Lukas Röd, Anna-Lena Eckert, Heiner Stuke, Andreas Heinz, and Philipp Sterzer. Psychotic experiences in schizophrenia and sensitivity to sensory evidence. *Schizophrenia Bulletin*, volume 46(4), pages 927–936, July 2020.

2019 Heiner Stuke, **Veith Weilnhammer**, Philipp Sterzer, and Katharina Schmack. Delusion proneness is linked to a reduced usage of prior beliefs in perceptual decisions. *Schizophrenia Bulletin*, volume 45(1), pages 80–86, January 2019.

2018 **Veith Weilnhammer**, Heiner Stuke, Philipp Sterzer, and Katharina Schmack. The neural correlates of hierarchical predictions for perceptual decisions. *The Journal of Neuroscience*, volume 38(21), pages 5008–5021, May 2018.

2017 **Veith Weilnhammer**, Heiner Stuke, Guido Hesselmann, Philipp Sterzer, and Katharina Schmack. A predictive coding account of bistable perception - a model-based fMRI study. *PLOS Computational Biology*, volume 13(5), page e1005536, May 2017.

2017 Heiner Stuke, Hannes Stuke, **Veith Weilnhammer**, and Katharina Schmack. Psychotic experiences and overhasty inferences are related to maladaptive learning. *PLOS Computational Biology*, volume 13(1), page e1005328, January 2017.

2016 **Veith Weilnhammer**, Philipp Sterzer, and Guido Hesselmann. Perceptual stability of the Lissajous figure is modulated by the speed of illusory rotation. *PLOS ONE*, volume 11(8), page e0160772, August 2016.

2016 Katharina Schmack, **Veith Weilnhammer**, Jakob Heinzle, Klaas E. Stephan, and Philipp Sterzer. Learning what to see in a changing world. *Frontiers in Human Neuroscience*, volume 10, page 263, May 2016.

2014 **Veith Weilnhammer**, Karin Ludwig, Philipp Sterzer, and Guido Hesselmann. Revisiting the Lissajous figure as a tool to study bistable perception. *Vision Research*, volume 98, pages 107–112, May 2014.

2013 **Veith Weilnhammer**, Karin Ludwig, Guido Hesselmann, and Philipp Sterzer. Frontoparietal cortex mediates perceptual transitions in bistable perception. *The Journal of Neuroscience*, volume 33(40), pages 16009–16015, October 2013.

Preprints

2026 **Veith Weilnhammer**, Lennart Luettgau, Christopher Summerfield, Raymond Dolan, Elise Wilkinson, Virginia Corno, Viknesh Sounderajah, and Matthew M. Nour. Single-turn emergency psychiatric triage across 15 frontier AI chatbots. *arXiv*, page 2604.25415 [q-bio.NC], April 2026.

2025 **Veith Weilnhammer**, Jefferson Ortega, and David Whitney. Human-computer interactions predict mental health. *arXiv*, page 2511.20179 [q-bio], November 2025.

Invited Presentations

2027 Investigating the mechanisms of mental health and illness at scale. *Talk*, Brain Meeting, Functional Imaging Laboratory, University College London, United Kingdom, 5 February (forthcoming).

2026 Representation learning from digital behavior. *Talk*, Behavioral Data Science, UCL Psychology, London, United Kingdom, 2 December (forthcoming).

2026 Using human-computer interactions to understand mental states linked to psychotic experiences. *Keynote*, International Consortium on Hallucination Research / ECHR Satellite Meeting, Technical University of Munich, Germany, 19 October.

2025 New ways to measure mental health. *Keynote*, MESEC Workshop, Lago di Bolsena, Tuscany, Italy.

2025 The geometry of mental health. *Talk*, UCL–DRCMR Symposium, Copenhagen, Denmark.

2023 Are psychotic symptoms linked to an overweighting of prior knowledge or of sensory information: Is there a consensus? *Symposium*, SIRS, Toronto, Canada.

2022 Computational modelling and non-invasive brain stimulation reveal an active role of inferior frontal cortex in resolving sensory ambiguity. *Symposium*, European Conference on Visual Perception, Nijmegen, Netherlands.

2021 Using visual illusions to understand the role of prefrontal cortex in conscious experience. *Talk*, NeuroSpin Conference, Saclay, France.

2021 Inferior frontal cortex as a potential target for non-invasive brain stimulation in the treatment of hallucinations. *Symposium*, DGPPN Annual Congress, Berlin, Germany.

2019 A predictive-coding account of altered perceptual inference in schizophrenia. *Symposium*, European Conference on Schizophrenia Research, Berlin, Germany.

2016 Hierarchical predictions for perceptual decisions. *Talk*, International Workshop on Schizophrenia and Predictive Coding, Marstrand, Sweden.

2015 Perceptual decisions in volatile environments. *Talk*, Highlands Meeting on Schizophrenia and Predictive Coding, Glencoe, Scotland.

Software

2025 **Veith Weilnhammer**. P2P Human Authenticator. https://pixelblot.ai, May 2025.

Individual Research Support

Ongoing and Awarded

2027–2029 **Authenticating Survey Responses Through Human–Computer Interactions**, Principal Investigator, Brain & Behavior Research Foundation, $70,000.

2026–2027 **Enhancing the safety of mental health chatbots through non-verbal user state inference**, Principal Investigator, UCL Inspire Innovation Fund, £20,000.

Completed

2025–2026 **Adversarial attacks on the human visual system**, Principal Investigator, Brain & Behavior Research Foundation, $35,000.

2023–2025 **Hallucinations in biological and artificial neural networks**, Fellow, Leopoldina – National Academy of Sciences, €100,800.

2019–2022 **The effect of prefrontal lesions on conscious experience**, Clinician Scientist Program, Berlin Institute of Health, 0.5 FTE.

2017–2018 **The influence of prior knowledge on perception in schizophrenia**, Junior Clinician Scientist Program, Berlin Institute of Health, 0.2 FTE.

2013 **Dynamic causal modelling of intracranial EEG**, Investigator, Minerva Foundation, Max Planck Society, €5,200.

Teaching

2021–2022 **Practical classes**: Cognitive Functions of the Frontal Cortex

2018–2022 **Lecture**: Computational Models of Schizophrenia

2018–2022 **Online lecture**: Schizophrenia as a Model Disorder

2016–2020 **Problem-based learning**: Basic and clinical medical modules

2015–2022 **Bedside teaching**: Biopsychosocial History and Psychopathology, mood disorders, adjustment disorders, psychotic disorders, cognitive disorders, liaison psychiatry

2015–2017 **Seminars and large-group teaching**: Mental Representations and Information Processing

Skills and Qualifications

Clinical Psychotherapy (cognitive behavioural, systemic, Open Dialogue); ketamine- and TMS-assisted treatments; deep brain stimulation

Research Machine learning and computational modelling in Python, R, Matlab, and JavaScript; psychophysics; fMRI; M/EEG; TMS; Neuropixels

Teaching Higher education teaching certifications in online teaching (2022) and problem-based learning (2015)

Research Supervision and Mentorship

2026–present **Supervision of postdoctoral research**, Lisa Kluen, University College London.

2026–present **Member of thesis committee**, Luna Huestegge, University College London.

2026–present **Supervision of research internship**, Nadim Benadrouche, Max Planck UCL Centre, London.

2026 **Supervision of undergraduate research project**, Kevin Hou, University College London.

2023–2025 **Supervision of undergraduate research project**, Jasmine Lopez, University of California, Berkeley.

2020–2025 **Co-supervision of MD thesis**, Katrin Reichenbach, Charité, Berlin.

2019 **Co-supervision of Master's thesis**, Deniz Yilmaz, Berlin School of Mind and Brain, Berlin.

2019 **Supervision of Master's thesis**, Meera Chikermane, NeuroCure, Berlin.

2018–2022 **Co-supervision of PhD thesis**, Anna-Lena Eckert, Berlin School of Mind and Brain, Berlin.

2018–2019 **Co-supervision of Master's thesis**, Drew Cooper, NeuroCure, Berlin.

2018–2019 **Co-supervision of Master's thesis**, Lukas Röd, Berlin School of Mind and Brain, Berlin.

2017–2025 **Co-supervision of postdoctoral research project**, Merve Fritsch, Charité, Berlin.

Memberships

2024–present **Association for the Scientific Study of Consciousness**

2015–present **Medical Association**, Berlin, Germany

Academic Service

Peer review Journal of Neuroscience (3), Communications Psychology (2), Journal of Vision (2), Schizophrenia Bulletin (2), Biological Psychiatry (1), Neuroscience of Consciousness (1), PLOS Computational Biology (1), PNAS (1), Trends in Cognitive Sciences (1), Neuropsychopharmacology (1)

Thesis review Monash University (1)

Grant review Deutsche Forschungsgemeinschaft (DFG) (1)

Public Engagement and Media

2026 **SPIEGEL Online**: Interview on SIM-VAIL and the psychological risks of AI chatbots, including their potential to reinforce delusional beliefs and exacerbate mental health vulnerabilities.